

Sequential Circuit

41

A flip-flop circuit can be used for:

- 1. counting
- 2. scaling
- 3. demodulation
- 4. More than one of the above

42

A flip-flop can store:

- 1. One bit of data
- 2. Two bits of data
- 3. Three bits of data
- 4. Any number of bits of data

43

A single flip-flop is a modulo _____ counter.

- 1. 1
- 2. 2
- 3. 0
- 4. More than one of the above

44

Race Around condition can be avoided in Digital logic circuits using?

- 1. Shift Register
- 2. Master-Slave JK Flip Flop
- 3. Full Adder
- 4. More than one of the above

Sequential Circuit

45

Which condition is shown in J-K flip flop as no changes next state from the present state?

1. $J = 0, K = 0$

2. $J = 0, K = 1$

3. $J = 1, K = 0$

4. $J = 1, K = 1$

46

D flip flop can be made from a J-K flip flop by making

1. $J = K$

2. $J = K = 1$

3. $J = 0, K = 1$

4. $J = \bar{K}$

47

Which is used for storing the one-bit digital data?

1. NAND GATE

2. GATE

3. Flip flop

4. Register

48

Master-slave configuration is used in FF to

1. increase its clocking rate

2. reduces power dissipation

3. eliminates race around condition

4. improves its reliability

Sequential Circuit

49

When two asynchronous active low inputs PRESET and CLEAR are applied to a J-K flip flop the output will be

- 1. 0
- 2. Undefined
- 3. Previous state
- 4. 1

50

Which of the following logic circuits do not have no-change condition?

- 1. D-FF
- 2. T-FF
- 3. JK-FF
- 4. SR-Latch

51

Which of the following is the other name for the D flip-flop?

- 1. Dual flip-flop
- 2. Decimal flip-flop
- 3. Delay flip-flop
- 4. Decay flip-flop

52

How many Flip flops circuits are needed to divide by 16?

- 1. Two
- 2. Four
- 3. Eight
- 4. Sixteen

Sequential Circuit

53

The one input RS flip flop is the _____ flip flop

1. T

2. D

3. R

4. Latch

54

If input to T flip flop is 200 Hz signal, then what will be the output signal frequency if four T flip flops are connected in cascade

1. 200 Hz

2. 50 Hz

3. 800 Hz

4. None of the above

$$f_{\text{out}} = \frac{\text{Input Frequency}}{2^n}$$

55

When both the inputs of a latch are high, the output is unpredictable. What is this condition called?

1. Bistable

2. Indeterminate

3. Inactive

4. No change

56

In S-R latch (NOR), when the SET input is made high, output Q becomes:

1. 0

2. 1

3. no change

4. application not allowed

Sequential Circuit

57

Which property is NOT considered in latches?

1. Output of the latches changes as we change the input.
2. changes as we change the input.
3. Latches are edge triggered.
4. Latches are fast.

58

_____ is commonly used to interface output devices.

1. Buffer
2. Pulse generator
3. Accumulator
4. Latch

59

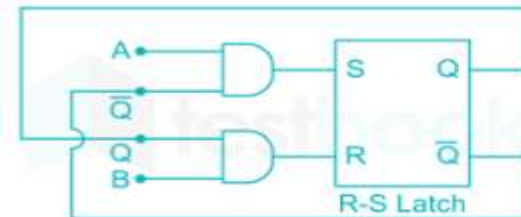
In the latch circuit shown, the NAND gates have non-zero, but unequal propagation delays. The present input conditions is: $P = Q = '0'$. If the input conditions is changed simultaneously to $P = Q = '1'$, the outputs X and Y are



1. $X = '1', Y = '1'$
2. either $X = '1', Y = '0'$ or $X = '0', Y = '1'$
3. either $X = '1', Y = '1'$ or $X = '0', Y = '0'$
4. $X = '0', Y = '0'$

60

The two inputs A and B are connected to a NOR based R-S latch, via two AND gates as shown in the figure. If $A = 1$ and $B = 0$, the output $Q\bar{Q}$ is

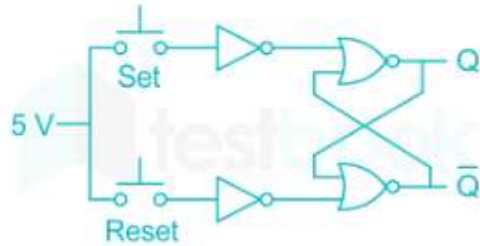


1. 00
2. 10
3. 01
4. 11

Sequential Circuit

61

An SR latch is implemented using TTL gates as shown in the figure. The set and reset pulse inputs are provided using the push-button switches. It is observed that the circuit fails to work as desired. The SR latch can be made functional by changing



1. NOR gates to NAND gates
2. inverters to buffers
3. NOR gates to NAND gates and inverters to buffers
4. 5 V to ground

62

Which of the following is true?

1. A flip-flop and latch both function same
2. flip-flop is edge triggered and latch is level triggered
3. Flip-flop is level triggered and latch is edge triggered
4. Either of flip-flop and latch is a combinational circuit

63

72). The BCD is one type of counter which is also known as _____

- ☐ Synchronous
- ☐ Asynchronous
- ☐ Parallel
- ☒ Decade

64

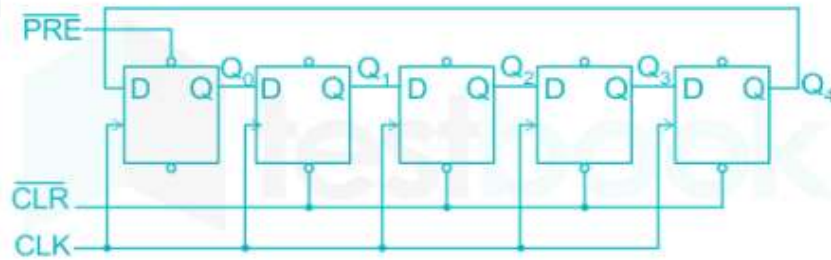
65). The flip flops are activated by _____ trigger

- ☐ Only positive edge
- ☐ Only negative edge
- ☒ Either positive or negative edge
- ☐ None of the above

Sequential Circuit

65

The following circuit is a



1. Mod-5 Ring counter
2. Mod-5 Johnson counter
3. Mod-10 Ring counter
4. Mod-10 Johnson counter

66

For the multistage counter arrangement of the given figure, determine the frequency of the output signal in Hz.



- a) 125Hz
- b) 1 MHz
- c) 1 KHz
- d) 125KHz

67

A traffic signal cycles from GREEN to YELLOW, YELLOW to RED and RED to GREEN. In each cycle, GREEN is turned on for 70 seconds, YELLOW is turned on for 5 seconds and the RED is turned on for 75 seconds. This traffic light has to be implemented using a finite state machine (FSM). The only input to this FSM is a clock of 5 second period. The minimum number of flip-flops required to implement this FSM is _____.

- a) 6
- b) 30
- c) 5
- d) None

68

2. What is the difference between a shift-right register and a shift-left register?

- a) There is no difference
- b) The direction of the shift
- c) Propagation delay
- d) The clock input

2. Digital Logic and Microprocessor

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Syllabus

2.1 Digital logic: Number Systems, Logic Levels, Logic Gates, Boolean algebra, Sum-of-Products Method, Product-of-Sums Method, Truth Table to Karnaugh Map. (AExE0201)

2.2 Combinational and arithmetic circuits: Multiplexetures, Demultiplexetures, Decoder, Encoder, Binary Addition, Binary Subtraction, operation on Unsigned and Signed Binary Numbers. (AExE0202)

2.3 Sequential logic circuit: RS Flip-Flops, Gated Flip-Flops, Edge Triggered Flip-Flops, Mater- Slave Flip-Flops. Types of Registers, Applications of Shift Registers, Asynchronous Counters, Synchronous Counters. (AExE0203)

2.4 Microprocessor: Internal Architecture and Features of microprocessor, Assembly Language Programming. (AExE0204)

2.5 Microprocessor system: Memory Device Classification and Hierarchy, Interfacing I/O and Memory Parallel Interface. Introduction to Programmable Peripheral Interface (PPI), Serial Interface, Synchronous and Asynchronous Transmission, Serial Interface Standards. Introduction to Direct Memory Access (DMA) and DMA Controllers. (AExE0205)

2.6 Interrupt operations: Interrupt, Interrupt Service Routine, and Interrupt Processing. (AExE0206)

2.4 Microprocessor: Internal Architecture and Features of microprocessor, Assembly Language Programming. (AExE0204)

INTRODUCTION TO MICROPROCESSOR

- Microprocessor is a multipurpose, programmable, clock-driven, register-base, electronic device that reads binary instructions from a storage device called memory, accepts binary data as input and processes data according to instructions, and provides results as output.
 - ❖ Input: Digital input from I/O or Memory
 - ❖ Process: Process the instruction
 - ❖ Output: Digital output to I/O or memory
- A typical programmable machine can be represented with four components: microprocessor, memory, input, and output. The physical components of this system are called hardware. A set of instructions written for the microprocessor to perform a task is called a program, and a group of program is called software.

INTRODUCTION TO MICROPROCESSOR

- In general microprocessor performs three basic tasks:
 1. Data transfer between itself and memory or I/O unit
 2. Arithmetical or Logical operation on data, and
 3. Switching and decision making
- Operation on above tasks are based on the instruction provided to the microprocessor. So, a critical steps in performing any task is retrieval of instruction, understanding the command and executing it.
- This can be listed as:
 1. Fetch the instruction
 2. Decode instruction
 3. Execute the command from the instruction

INTRODUCTION TO MICROPROCESSOR

- A microprocessor incorporates the functions of a computer's central processing unit (CPU) on a single integrated chip (IC).

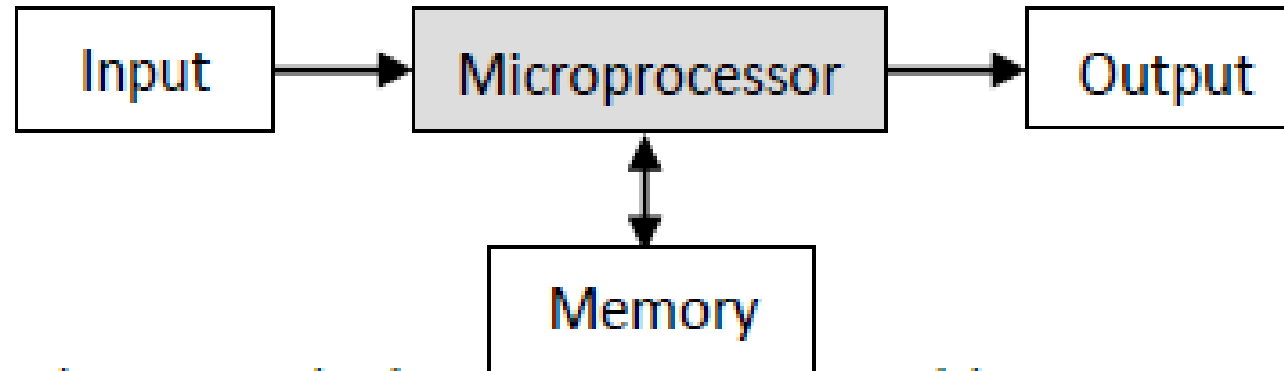


Figure: typical programmable machine

MICROCONTROLLER

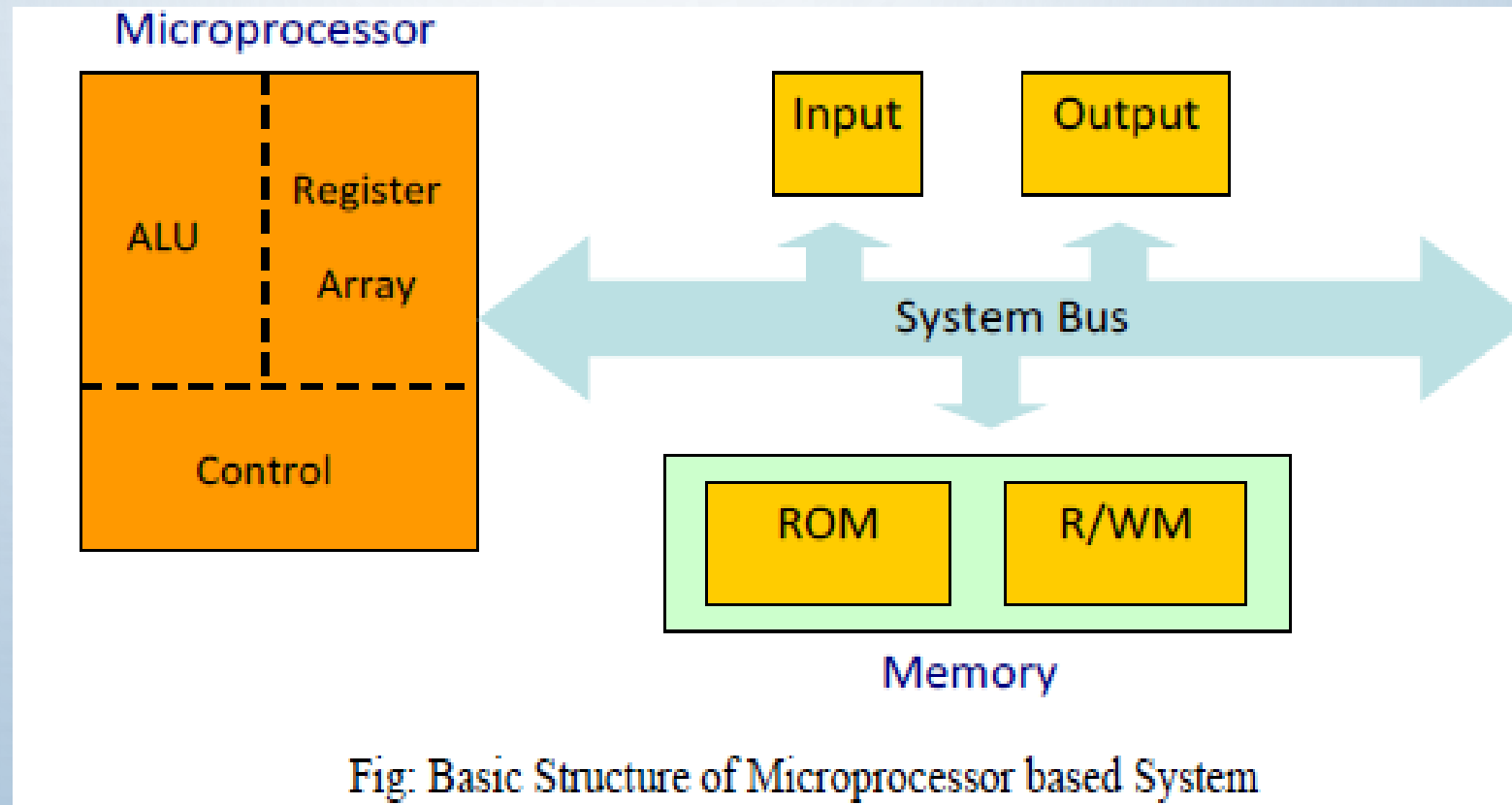
- A microcontroller is a small computer on a single integrated circuit containing a processor core, memory, and programmable input/output peripherals. Microcontrollers are designed for embedded applications, in contrast to the microprocessors used in personal computers or other general purpose applications.

MICROPROCESSOR VS MICROCONTROLLER

Microprocessor	Micro Controller
 <pre> graph LR MP[Microprocessor] <--> System Bus ROM[Read-Only Memory (ROM)] MP <--> System Bus RWM[Read-Write Memory] MP <--> System Bus T[Timer] MP <--> System Bus IO[I/O Port] MP <--> System Bus SI[Serial Interface] </pre>	 <pre> graph LR subgraph MC [Microcontroller] ROM[Read-Only Memory] RWM[Read-Write Memory] T[Timer] IO[I/O Port] SI[Serial Interface] end </pre>
Microprocessor is heart of Computer system.	Micro Controller is a heart of embedded system.
It is just a processor. Memory and I/O components have to be connected externally	Micro controller has external processor along with internal memory and i/o components
Since memory and I/O has to be connected externally, the circuit becomes large.	Since memory and I/O are present internally, the circuit is small.
Cannot be used in compact systems and hence inefficient	Can be used in compact systems and hence it is an efficient technique
Cost of the entire system increases	Cost of the entire system is low
Due to external components, the entire power consumption is high. Hence it is not suitable to used with devices running on stored power like batteries.	Since external components are low, total power consumption is less and can be used with devices running on stored power like batteries.
Most of the microprocessors do not have power saving features.	Most of the micro controllers have power saving modes like idle mode and power saving mode. This helps to reduce power consumption even further.
Since memory and I/O components are all external, each instruction will need external operation, hence it is relatively slower.	Since components are internal, most of the operations are internal instruction, hence speed is fast.
Microprocessor have less number of registers, hence more operations are memory based.	Micro controller have more number of registers, hence the programs are easier to write.
Microprocessors are based on von Neumann model/architecture where program and data are stored in same memory module	Micro controllers are based on Harvard architecture where program memory and Data memory are separate
Mainly used in personal computers	Used mainly in washing machine, MP3 players

ORGANIZATION OF MICROPROCESSOR BASED SYSTEM

- The basic structure of microprocessor based system includes microprocessor, I/O and memory (ROM & R/WM). These components are organized around a common communication path called bus.



BUS ORGANIZATION/ 3-BUS ARCHITECTURE

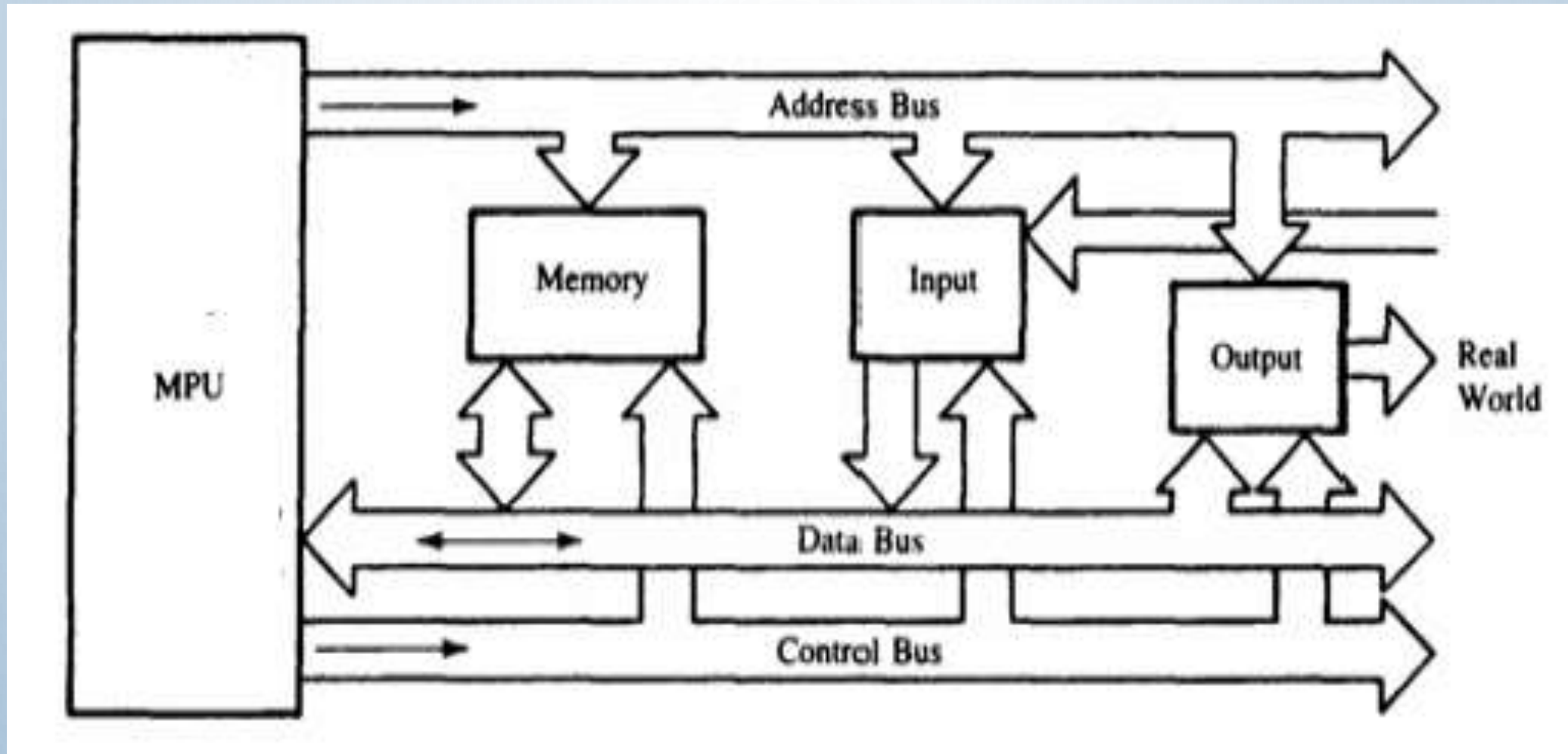


Fig: 3-Bus Architecture of Microprocessor

BUS ORGANIZATION/ 3-BUS ARCHITECTURE

Address bus:

- The address bus consists of 16, 20, 24 or 32 parallel signal lines that is used to specify a physical address.
- On these lines the CPU sends out the address of the memory location that is to be written to or from.
- The address bus is unidirectional: bit flow in one direction- from the MPU to peripheral devices.
- The width of the address bus determines the amount of memory a system can address.
- For example, a system with a 16-bit address bus can address 2^{16} (64 KB) memory locations.

BUS ORGANIZATION/ 3-BUS ARCHITECTURE

Data bus:

- The data bus consist of 8, 16 or 32 parallel signal lines and are bidirectional that carries the actual data being processed.
- CPU can read data in from memory and send data out to memory on these lines.

BUS ORGANIZATION/ 3-BUS ARCHITECTURE

Control bus:

- The control bus is comprised of various single lines that carry synchronization signals.
- The MPU uses such lines to provide timing signals.
- These are no a group of lines like data and address buses, but individual lines that provide a pulse to indicate an MPU operation.
- The MPU generates specific control signals for every operations (such as Memory read or I/O write) it performs.
- These signals are used to identify a device type with which the MPU intended to communicate.

STORED PROGRAM CONCEPT AND VON NEUMANN MACHINE

- The task of entering and altering the programs for the ENIAC (electronic numerical integrator and computer) was extremely tedious.
- The programming concept could be facilitated if the program could represent in a form suitable for storing in memory alongside the data.
- Then a computer could get its instruction by reading them from the memory and a program could be set or altered by setting the values of a portion of memory.
- This approach is known “stored program concept”, was first adopted by John Von Neumann and hence the architecture of computer he proposed is name as Von-Neumann’s architecture.
- (Note: 8085 μ p uses this architecture)

STORED PROGRAM CONCEPT AND VON NEUMANN MACHINE

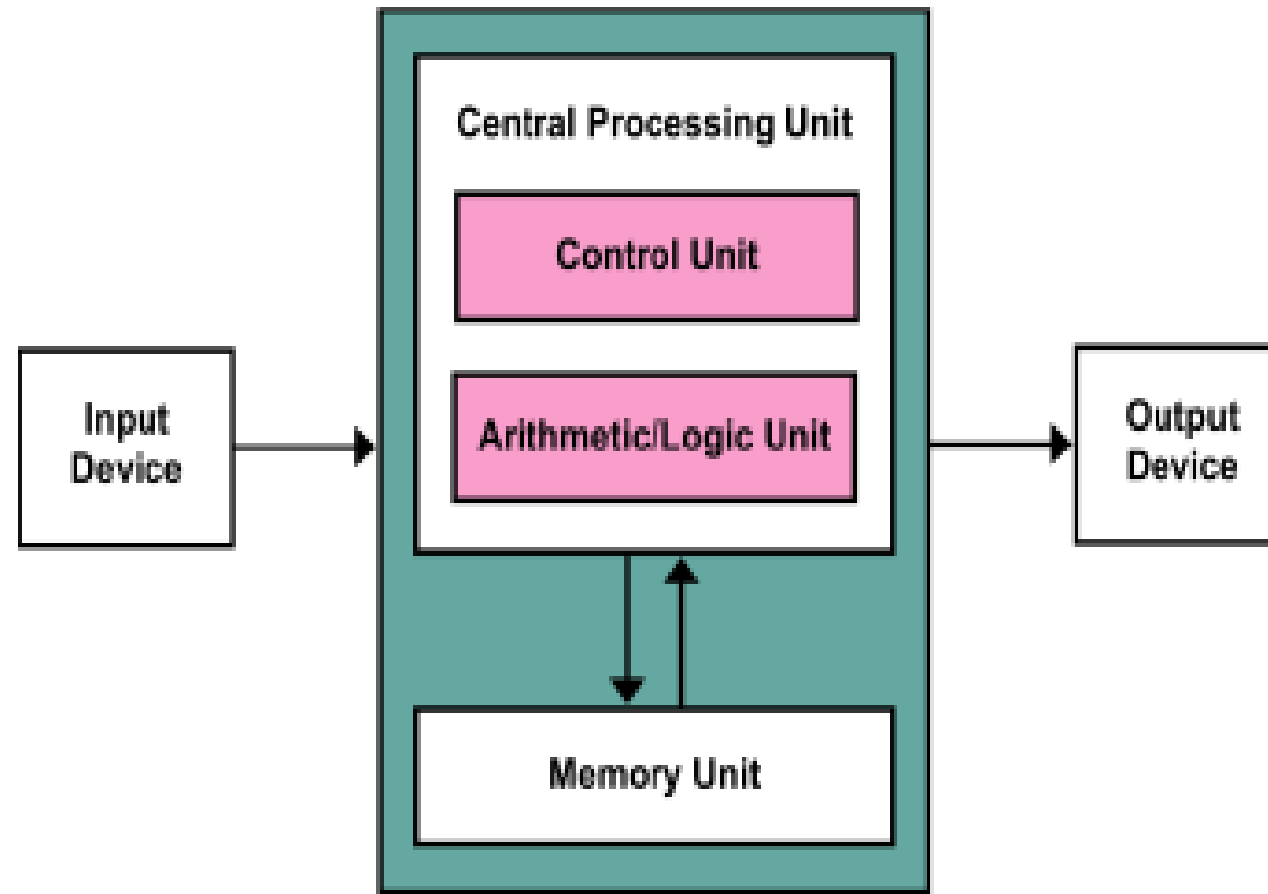


Fig: Von-Neumann Architecture

STORED PROGRAM CONCEPT AND VON NEUMANN MACHINE

- Main memory is used to store both data and instructions.
- The ALU is capable for performing arithmetic and logical operation on binary data.
- The control unit (CU) interprets the instruction in memory and causes them to be executed.
- The I/O unit gets operated from the control unit. The input/output unit helps inputting data and getting results.
- The Von-Neumann's Architecture is the fundamental basis for the architecture of today's digital computers.

STORED PROGRAM CONCEPT AND VON NEUMANN MACHINE

- The memory of Von-Neumann machine consists of thousand storage location called words of 40 binary digits (bits).
- Both data and instruction are stored in it.
- The storage locations of control unit and ALU are called registers.
- The various registers of this model are MBR, MAR, IR, IBR, PC, AC.
- **Memory Buffer Register (MBR):**
 - It consists of a word to be stored in memory or is used to receive a memory or is used to receive a word from memory.
- **Memory address Register (MAR):**
 - It contains the address in memory of the word to be written from or read into the MBR.

STORED PROGRAM CONCEPT AND VON NEUMANN MACHINE

- **Instruction register (IR):**

- Contain the 8 bit op-code (operation code) instruction being executed.

- **Instruction buffer register (IBR):**

- It is used to temporarily hold the instruction from a word in memory.

- **Program counter (PC):**

- It contains address of next instruction to be fetched from memory.

- **Ac (Accumulator) and MQ (multiplier quotient):**

- They are employed to temporarily hold operands and results of ALU operations.

STORED PROGRAM CONCEPT AND VON NEUMANN MACHINE

Advantages:

- Computer can handle instruction as easily as data
- Ease of loading program into memory
- Efficient use of memory
- Cost effective due to same program and data memory

Disadvantages:

- Required special hardware protection mechanism to protect instruction and data being overlapped by each other.
- Low speed because concurrent fetching of data and instruction was not possible.

HARVARD ARCHITECTURE

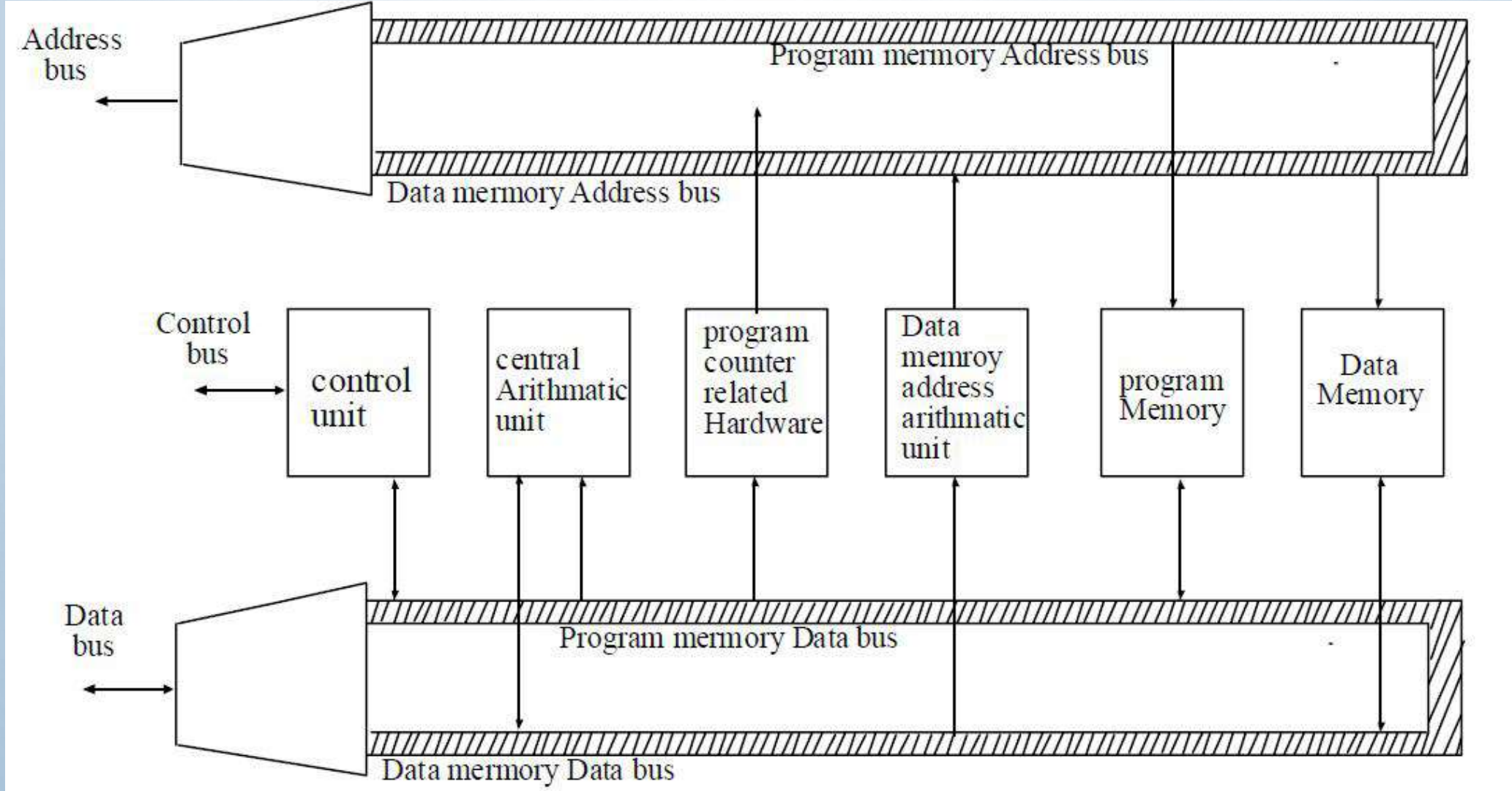


Fig. Block diagram of the Harvard architecture

HARVARD ARCHITECTURE

- Harvard Architecture based computer consist so separate memory spaces for the programs (instruction) and data.
- Each memory space has its own address and data bus.
- Thus both instruction and data can fetch from memory concurrently.
- From the figure it is seen that there are two data and two address buses for the program and data memory spaces respectively.
- The program memory data bus and data memory data are multiplexed to form single data bus where as program memory Address and data memory address are multiplexed to form single address bus.

HARVARD ARCHITECTURE

- Hence there are two blocks of RAM chip: One for program memory and another for data memory space.
- Data memory address arithmetic unit generates data memory address.
- The data memory address bus carries the memory address of data where as program memory address bus carries the memory address of the instruction.
- Central arithmetic logic unit consists of the ALU, multiplier, Accumulator, etc.
- The Program Counter is used to address program memory.
- PC always contains the address of next instruction to be fetched. Control unit control the sequence of operations to be executed.
- The data and control bus are bidirectional where as address bus is unidirectional.

HARVARD ARCHITECTURE

Advantages:

- Concurrent fetching of data and instruction was possible so it provide higher speed
- No overwriting of program and data

Disadvantages:

- Methods or mechanism of storing program into program memory and data into data memory had to be developed
- Higher cost due to separate program and data memory
- No optimum use of memory.

INTRODUCTION TO REGISTER TRANSFER LANGUAGE (RTL)

FETCH – REGISTERS

- **Memory Address Register (MAR)**
 - Connected to address bus
 - Specifies address for read or write op
- **Memory Buffer Register (MBR)**
 - Connected to data bus
 - Holds data to write or last data read
- **Program Counter (PC)**
 - Holds address of next instruction to be fetched
- **Instruction Register (IR)**
 - Holds last instruction fetched

INTRODUCTION TO 8085 MICROPROCESSOR

- The 8085 μ p is an 8 bit general purpose microprocessor having 16 bit address lines (capable of addressing $2^{16} = 65536$ bytes = 64 KB of memory).
- The device has 40 pins, requires a +5V single power supply and can operate with 3 MHZ single phase clock.

INTERNAL ARCHITECTURE AND FEATURE OF 8085 MICROPROCESSOR

- The main components of 8085 μ p, as shown in functional block diagram, are the arithmetic/logic unit (ALU), Register array, timing and control unit, instruction register and decoder, interrupt control and serial I/O control. These are linked by an internal data bus.

INTERNAL ARCHITECTURE AND FEATURE OF 8085 MICROPROCESSOR

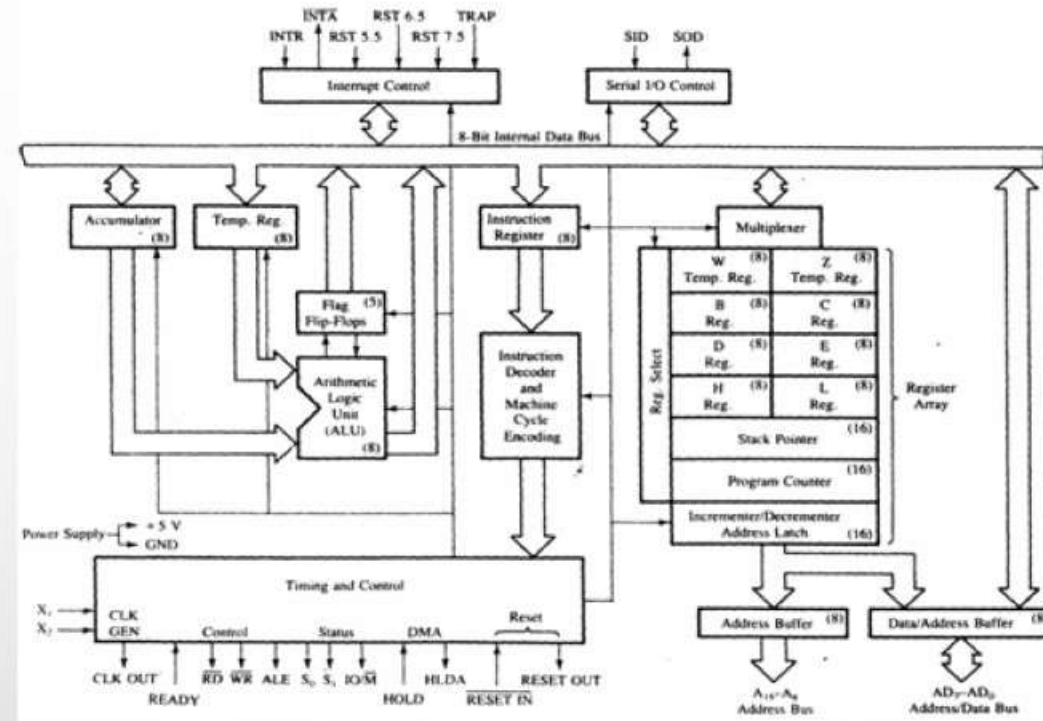
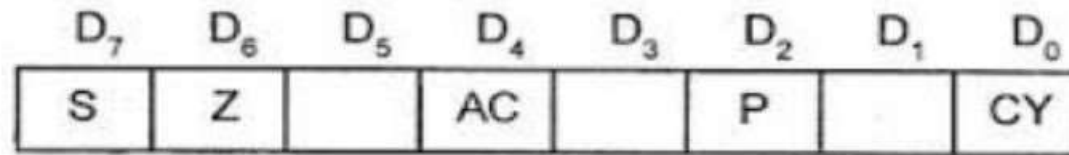


Figure: functional block diagram of 8085 microprocessor

INTERNAL ARCHITECTURE AND FEATURE OF 8085 MICROPROCESSOR



➤ S- Sign flag:

- After the execution of arithmetic or logic operation, if D₇ bit (MSB: most significant bit) of the result (usually in accumulator) is 1, the sign flag (S) is set. Otherwise it is reset. (For signed number operation D₇ bit indicates the sign(positive or negative) of number).

➤ Z- Zero flag:

- Set i.e. 1, if the result of last operation is zero, and the flag is reset i.e. 0 if the result is not 0. This flag is often used in loop control and in searching for particular data value.

INTERNAL ARCHITECTURE AND FEATURE OF 8085 MICROPROCESSOR

➤ **AC- Auxiliary Carry:**

- In an arithmetic operation, when a carry is generated by digit D3 and passed on to digit D4, the AC flag is set. The flag is used only internally for BCD operations and is not available for programmer.

➤ **P- Parity flag:**

- Set i.e. 1, if the number of 1 in the result of the last operation is even. If the result has an odd number of 1's the flag is reset i.e. 0.

➤ **C- Carry flag:**

- Flag is set i.e. 1, if the result of the last operation generates a carry, otherwise it is reset i.e. 0.

INTERNAL ARCHITECTURE AND FEATURE OF 8085 MICROPROCESSOR

Instruction register and decoder :

- The instruction register receives the operation codes of instructions from the internal data bus and passes it to the instruction decoder and machine cycle encoder circuit.
- The decoder decodes the instruction and establishes the sequence of events to follow.
- The instruction register is not accessible to the programmer.

Register array :

- The 8085 μ p has 6 general purpose registers to hold 8 bit data; these are B, C, D, E, H, and L.
- They can also be combined as register pairs BC, DE, And HL to perform 16 bit operation.
- These registers are accessible to programmer i.e. programmable.
- In addition, the register H and L are utilized in indirect addressing mode. In this mode, the memory location whose address is specified by contents of the register pair.

INTERNAL ARCHITECTURE AND FEATURE OF 8085 MICROPROCESSOR

- Register array also includes two additional register W & Z, called temporary registers, each used to hold 8 bit data during the execution of some instructions.
- However, they are not available to programmer.
- Program counter (PC) and Stack Pointer (SP) are two 16 bit registers used to hold memory addresses.
- A stack is an area of R/W memory set aside for the purpose of storing data, by an operation known as stacking.
- The beginning of stack is defined by loading a 16 bit address in the stack pointer register.
- A computer program consists of the sequence of coded instructions. These instructions are stored sequentially in the memory location.

INTERNAL ARCHITECTURE AND FEATURE OF 8085 MICROPROCESSOR

Interrupt controls:

- The various interrupt controls signals (INTR, RST 5.5, RST 6.5, RST 7.5 and TRAP) are used to interrupt a microprocessor.

Serial I/O controls:

- Two serial I/O control signals (SID and SOD) are used to implement the serial data transmission.

INTERNAL ARCHITECTURE AND FEATURE OF 8085 MICROPROCESSOR

The salient features of 8085 μ p are:

- It is a 8 bit microprocessor.
- It is manufactured with N-MOS technology.
- It has 16-bit address bus and hence can address up to $2^{16} = 65536$ bytes (64KB) memory locations through A_0 - A_{15} .
- The first 8 lines of address bus and 8 lines of data bus are multiplexed $AD_0 - AD_7$.
- Data bus is a group of 8 lines $D_0 - D_7$.
- It supports external interrupt request.

INTERNAL ARCHITECTURE AND FEATURE OF 8085 MICROPROCESSOR

- A 16 bit program counter (PC)
- A 16 bit stack pointer (SP)
- Six 8-bit general purpose register arranged in pairs: BC, DE, HL.
- It requires a signal +5V power supply and operates at 3 MHZ single phase clock.
- It is enclosed with 40 pins DIP (Dual in line package).
- It has 74 operation codes with total 246 instructions.

INSTRUCTION FORMAT

- An instruction is a binary pattern designed inside a μ p to perform a specific function.
- The entire group of instructions, called the instruction set.
- Each instruction has two parts: one is the task to be performed, called the operation code (Op-code), and the other is data to be operated on, called the operand.
- Operand may include 8 bit or 16 bit data, and internal register, a memory location, or an 8 (or 16 bit) address.
- In some instructions, the operand is implicit.
- The 8085 instruction set is classified into 3 groups according to word or byte size.

INSTRUCTION FORMAT

- 1- byte instructions
- 2- byte instructions
- 3- byte instructions

One byte instructions

- It includes the op-code and operand in the same byte. These instructions are stored in 8 bit binary format in memory; each requires one byte memory location.

e.g.

<u>Memory address</u>	<u>Op-code</u>	<u>Operand</u>	<u>Hex code</u>	<u>Description</u>
3000	MOV	C,A	4F H	Copy the contents of the Accumulator in register C
3001	ADD	B	80 H	Add the contents of register B to the contents of the Accumulator
3002	CMA		2F H	Invert (compliment) each bit in the Accumulator

INSTRUCTION FORMAT

Two byte instructions

- In this type of instruction, first byte specifies the operation code and the second byte specified the operand.
- These instructions would require two memory locations each to store the binary codes.

e.g.

<u>Memory address</u>	<u>Op-code</u>	<u>Operand</u>	<u>Hex code</u>	<u>Description</u>
3100	MVI	A, 32 H	3E H	Load an one byte data (32H) in the accumulator
3101			32 H	
8080	MVI	B, F2 H	06 H	Load an one byte data (F2H) in reg. B
8081			F2 H	

INSTRUCTION FORMAT

Three byte instructions

- In 3 byte instructions, the first byte specifies the Op-code, and the following two bytes specify the 16 bit address or data. (Note: second byte is the low-order address and the third byte is the high-order address).
- These instructions would require 3 memory locations each to store the binary codes.

e.g.

<u>Memory address</u>	<u>Op-code</u>	<u>Operand</u>	<u>Hex code</u>	<u>Description</u>
3000	LDA	2050 H	3A H	Load contents of memory 2050H into accumulator
3001			50 H	
3002			20 H	
8000	JMP	2085 H	C3 H	Transfer the program sequence to memory location 2085H
8001			85 H	
8002			20 H	

ADDRESS MODES OF 8085

- To perform any operation, we have to give the corresponding instructions to the microprocessor.
- In each instruction, we have to specify the following three things:
 - Operation to be performed.
 - Address of source of data.
 - Address of destination of result.
- The method by which the address of source of data or the address of destination of result is given in the instruction is called Addressing Mode.
- The term addressing mode refers to the way in which the operand of the instruction is specified.

ADDRESS MODES OF 8085

Types of Addressing Modes

- Intel 8085 uses the following addressing modes:
 - 1. Direct Addressing Mode
 - 2. Register Direct Addressing Mode
 - 3. Register Indirect Addressing Mode
 - 4. Immediate Addressing Mode
 - 5. Implied Addressing Mode

ADDRESS MODES OF 8085

Direct Addressing Mode

- Instructions using this mode specify the effective address as part of instruction.
- Instructions using this mode may contain 2 or 3 bytes, with first byte as the Op-code followed by 1 or 2 bytes or address of data.
- In this mode, the address of the operand is given in the instruction itself.

LDA 2500 H Load the contents of memory location 2500 H in accumulator.

LDA is the operation. 2500 H is the address of source. Accumulator is the destination.

IN 41 H Reads the data at port 41 H and store data at the accumulator

IN is the operation. 41 H is the address of source. Accumulator is the destination.

ADDRESS MODES OF 8085

Register Direct Addressing Mode

➤ In this mode, the operand is in general purpose register i.e. data is provided through registers.

➤ **MOV A, B** Move the contents of register B to A.

MOV is the operation. B is the source of data. A is the destination.

Register Indirect Addressing Mode

➤ In this mode, the address part of instruction specifies the memory location whose content is the address of the operand. In 8085 μ p, wherever the instruction uses the HL pointer the address is called indirect addressing.

➤ **MOV A, M** Move data from memory location specified by H-L pair to accumulator.

MOV is the operation. M is the memory location specified by H-L register pair. A is the destination.

LDAX B, STAX D

ADDRESS MODES OF 8085

Immediate Addressing Mode

- In this mode, the operand is specified within the instruction itself. This mode of instructions uses first byte as the Op-code and following 1 or 2 byte data itself.
 - MVI A, 05 H Move 05 H in accumulator.
MVI is the operation. 05 H is the immediate data (source). A is the destination.
 - LXI H, 7A21 H Loads register H with 7A H and register L with 21 H
LXI is the operation. 7A21 H is the immediate data (Source). H & L registers are the destination.
- So in both cases, the actual data is part of the instruction, and hence called immediate addressing.

ADDRESS MODES OF 8085

Implied Addressing Mode

- If address of source of data as well as address of destination of result is fixed, then there is no need to give any operand along with the instruction.
- So, instructions of such mode don't have operands.
 - CMA Complement accumulator.
CMA is the operation. A is the source. A is the destination.

REGISTERS IN 8085

Registers in 8085

- General purpose Register
- Temporary Register
- Special purpose Register
- 16 bit Register
- **General purpose register**
 - B, C, D, E, H, L are general purpose register They can store 8 bit of data at a time
 - 16 bit of data can also be stored by using combination of BC, DE and HL
 - The general purpose register is available for the user to manipulate

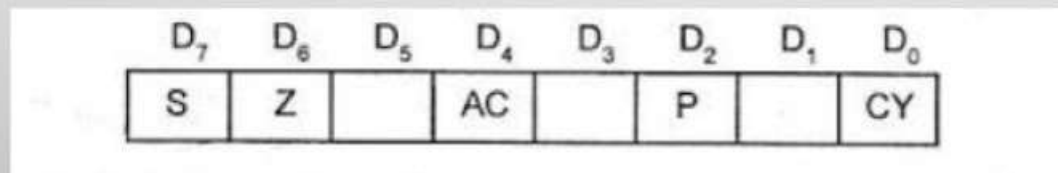
REGISTERS IN 8085

➤ Temporary Register

- W and Z are each of 8 bit and are used by processor for its internal operation and not available to the user or programmer

➤ Special purpose register

- Accumulator and flag are the two special purpose register each of 8 bit.
- Accumulator is extensively used to store results of an arithmetic and logical operation It is also used in input output operations.
- Flag register holds the properties of an arithmetical and logical operation



REGISTERS IN 8085

➤ 16 bit register

- Stack/ stack pointer and program counter are 16 bit registers
- Stack stores the value of flag and program counter during interrupt operation or subroutine call
- Stack pointer holds the address of top of the stack
- Program counter stores the address of next instruction to be executed

PROGRAMMING MODEL OF 8085

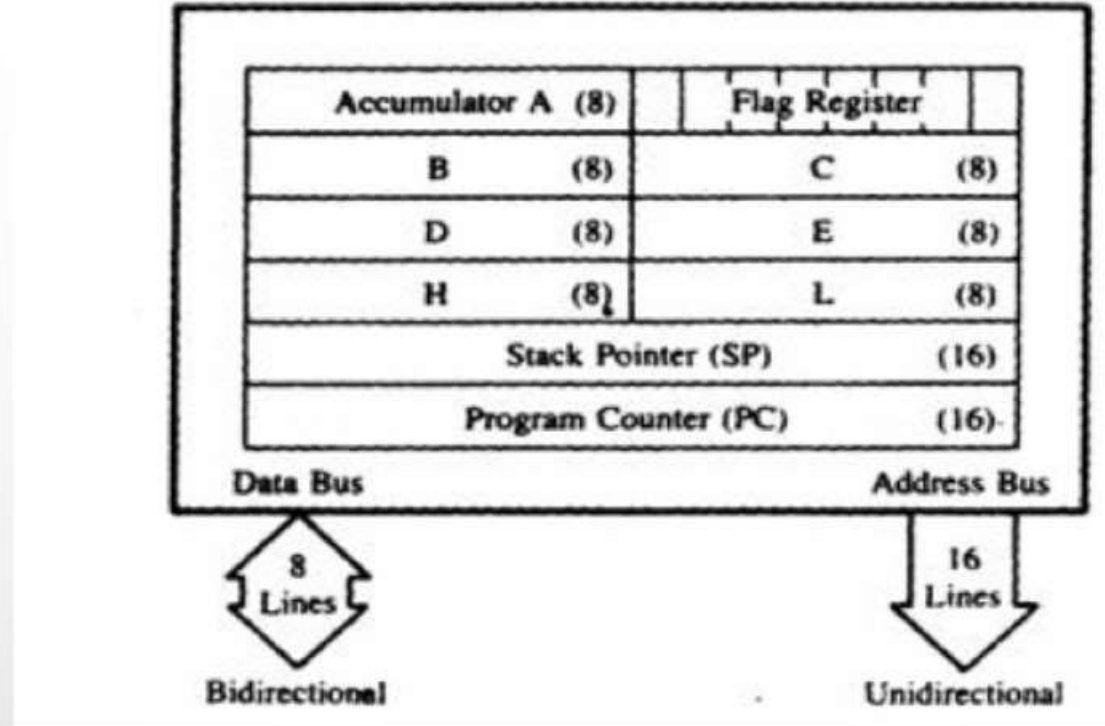


Fig: Programming model of 8085

Note: Explain Registers (Flags, Accumulator, Temporary Register and Register pair), Stack Pointer, Program Counter, Address and Data Bus

Introduction to 8086 Microprocessor

- 8086 Microprocessor is an enhanced version of 8085 Microprocessor that was designed by Intel in 1978.
- It is a 16 bit Microprocessor having 20 address lines and 16 data lines that provides up to 1 MB storage.
- It consists of powerful instruction set, which provides operations like multiplication and division easily.

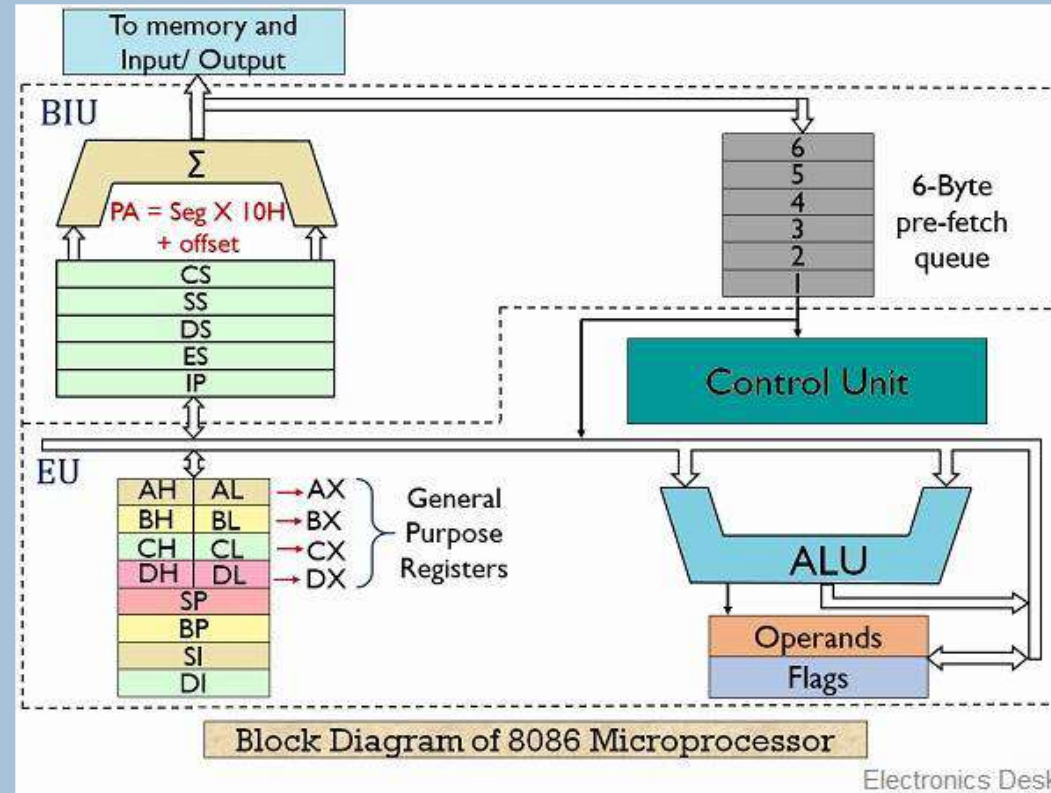
Features 8086 Microprocessor

- It has an instruction queue, which is capable of storing six instruction bytes from the memory resulting in faster processing
- It was the first 16 bit processor having 16 bit ALU, 16 bit registers, internal data bus, and 16 bit external data bus resulting in faster processing
- It is available at different clock frequencies 5 MHz, 8 MHz and 10 MHz
- It uses two stages of pipelining, i.e. Fetch Stage and Execute Stage, which improves performance
- Fetch stage can prefetch up to 6 bytes of instructions and stores them in the queue
- Execute stage executes these instructions
- It has 256 vectored interrupts
- It consists of 29 000 transistors

Comparison with 8085

- **Size:** 8085 is 8 bit microprocessor, whereas 8086 is 16 bit microprocessor
- **Address Bus :** 8085 has 16 bit address bus while 8086 has 20 bit address bus
- **Memory:** 8085 can access up to 64 KB, whereas 8086 can access up to 1 MB of memory
- **Instruction Queue:** 8085 doesn't have an instruction queue, whereas 8086 has an instruction queue
- **Pipelining:** 8085 doesn't support a pipelined architecture while 8086 supports a pipelined architecture
- **I/O:** 8085 can address $2^8 = 256$ I/O's, whereas 8086 can access $2^{16} = 65\,536$ I/O's
- **Cost:** The cost of 8085 is low whereas that of 8086 is high

Internal Architecture of 8086



Execution unit (EU) and its components

➤ Index register

- The two index registers SI (Source index) and DI (Destination Index) are used in indexed addressing. The instructions that process data strings use the SI and DI index register together with DS and ES respectively, in order to distinguish between the source and destination address.

➤ Flag register

- The 8086 has nine 1 bit flags. Out of 9 six are status and three are control flags. The control bits in the flag register can be set or reset by the programmer.

X	X	X	X	O	D	I	T	S	Z	X	Ac	X	P	X	Cy
---	---	---	---	---	---	---	---	---	---	---	----	---	---	---	----

Execution unit (EU) and its components

Control flags:

➤ **D-Direction Flag**

- This is used by string manipulation instructions. If this flag bit is '0' , the string is processed beginning from the lowest address to the higher address, i.e. auto incrementing mode otherwise the string is processed from the highest address towards the lowest address, i.e. autodecrementing mode.

➤ **I-Interrupt flag**

- If this flag is set the maskable interrupts are recognized by the CPU, otherwise they are ignored.

➤ **T- Trap flag**

- If this flag is set the processor enters the single step execution mode. In other words, a trap interrupt is generated after execution of each instruction. The processor executes the current instruction and the control is transferred to the Trap interrupt service routine.

Execution unit (EU) and its components

Status Flags:

➤ **O- Overflow flag**

- This flag is set if an arithmetic overflow occurs, i.e. if the result of a signed operation is large enough to be accommodated in a destination register.

➤ **S - Sign flag**

- This flag is set when the result of any computation is negative. For signed computations, the sign flag equals the MSB of the result.

➤ **Z- Zero**

- This flag is set when the result of the computation is or comparison performed by the previous instruction is zero. 1 for zero result, 0 for nonzero result

Execution unit (EU) and its components

➤ **A_c- Auxiliary Carry**

- This is set if there is a carry from the lowest nibble, i.e. bit three during the addition or borrow for the lowest nibble i.e. bit three, during subtraction.

➤ **P- Parity flag**

- This flag is set to 1 if the lower byte of the result contains even number of 1s otherwise reset.

➤ **C_y-Carry flag:**

- This flag is set when there is a carry out of MSB in case of addition or a borrow in case of subtraction.

Segment and offset address

- Segments are special areas defined in a program for containing the code, data and stack.
- A segment begins on a paragraph boundary.
- A segment register is of 16 bits in size and contains the starting address of a segment.
- A segment begins on a paragraph boundary, which is an address divisible by decimal 16 or hex 10.
- Consider a DS that begins at location 038E0H. In all cases, the rightmost hex digit is zero, the computer designers decided that it would be unnecessary to store the zero the zero digit in the segment register.
- Thus 038E0H is stores in register as 038EH.

Segment and offset address

- The distance in bytes from the segment address to another location within the segment is expressed as an offset or displacement.
- Suppose the offset of 0032H for above example of data segment. Processor combines the address of the data segment with the offset as:
- SA: OA (segment address: offset address)
- $038E: 0032\text{ H} = 038E * 10 + 0032 = 038E0 + 0032$
- Physical address = 03912H

Addressing modes in 8086

- Addressing modes describe types of operands and the way in which they are accessed for executing an instruction.
- An operand address provides source of data for an instruction to process an instruction to process.
- An instruction may have from zero to two operands. For two operands first is destination and second is source operand.
- The basic modes of addressing are register, immediate and memory which are described below.

Addressing modes in 8086

Register Addressing:

- For this mode, a register may contain source operand, destination operand or both.
- E.g. `MOV AH, BL` `MOV DX, CX`

Immediate Addressing

- In this type of addressing, immediate data is a part of instruction, and appears in the form of successive byte or bytes.
- This mode contains a constant value or an expression.
- E.g. `MOV AH, 35H` `MOV BX, 7A25H`

Addressing modes in 8086

Direct memory addressing:

- In this type of addressing mode, a 16-bit memory address (offset) is directly specified in the instruction as a part of it.
- One of the operand is the direct memory and other operand is the register.
- E.g. `ADD AX, [5000H]`
- Note: Here data resides in a memory location in the data segment, whose effective address may be computed using 5000H as the Offset address and content of DS as segment address.
- The effective address, here, is $10H * DS + 5000H$.

Addressing modes in 8086

Direct offset addressing

- In this addressing, a variation of direct addressing uses arithmetic operators to modify an address.
- E.g. `ARR DB 15, 17, 18, 21`
- `MOV AL, ARR [2]` ; `MOV AL, 18`
- `ADD BH, ARR+3` ; `ADD BH, 21`

Addressing modes in 8086

Indirect memory addressing:

- Indirect addressing takes advantage of computer's capability for segment: offset addressing.
- The registers used for this purpose are base register (BX and BP) and index register (DI and SI)

E.g. `MOV [BX],AL`

`ADD CX, [SI]`

Addressing modes in 8086

Base displacement addressing:

- This addressing mode also uses base registers (BX and BP) and index register (SI and DI), but combined with a displacement (a number or offset value) to form an effective address.
- E.g. `MOV BX, OFFSET ARR ;` `LEA BX, ARR`
- `MOV AL, [BX +2]`
- `ADD TBL [BX], CL` ;TBL [BX] [BX + TBL] e.g. [BX + 4]

Addressing modes in 8086

Base index addressing:

- This addressing mode combines a base registers (BX or BP) with an index register (SI or DI) to form an effective address.
- E.g. MOV AX, [BX +SI]
- ADD [BX+DI], CL

Base index with displacement addressing

- This addressing mode, a variation on base- index combines a base register, an index register, and a displacement to form an effective address.
- E.g. MOV AL, [BX+SI+2]
- ADD TBL [BX +SI], CH

Addressing modes in 8086

String addressing:

- This mode uses index registers, where SI is used to point to the first byte or word of the source string and DI is used to point to the first byte or word of the destination string, when string instruction is executed.
- The SI or DI is automatically incremented or decremented to point to the next byte or word depending on the direction flag (DF).
- E.g. MOVS, MOVSB, MOVSW

Coding in assembly language

- Assembly language programming has taken its place in between the machine language (low level) and the high level language.
 - High level language's one statement may generate many machine instructions.
 - Low level language consists of either binary or hexadecimal operation. One symbolic statement generates one machine level instructions.

Coding in assembly language

➤ **Advantage of ALP**

- They generate small and compact execution module.
- They have more control over hardware.
- They generate executable module and run faster.

➤ **Disadvantages of ALP**

- Machine dependent.
- Lengthy code
- Error prone (likely to generate errors).

Coding in assembly language

S.NO	HIGH LEVEL LANGUAGE	LOW LEVEL LANGUAGE
1.	It is programmer friendly language.	It is a machine friendly language.
2.	High level language is less memory efficient.	Low level language is high memory efficient.
3.	It is easy to understand.	It is tough to understand.
4.	It is simple to debug.	It is complex to debug comparatively.
5.	It is simple to maintain.	It is complex to maintain comparatively.
6.	It is portable.	It is non-portable.
7.	It can run on any platform.	It is machine-dependent.
8.	It needs compiler or interpreter for translation.	It needs assembler for translation.
9.	It is used widely for programming.	It is not commonly used now-a-days in programming.

assembly language features

- The main features of ALP are program comments, reserved words, identifies, statements and directives which provide the basic rules and framework for the language.

Program comments:

- The use of comments throughout a program can improve its clarity.
- It starts with semicolon (;) and terminates with a new line.
- E.g. `ADD AX, BX` ; Adds AX & BX

assembly language features

Reserved words

- Certain names in assembly language are reserved for their own purpose to be used only under special conditions and includes
 - **Instructions** : Such as MOV and ADD (operations to execute)
 - **Directives**: Such as END, SEGMENT (information to assembler)
 - **Operators**: Such as FAR, SIZE
 - **Predefined symbols**: such as @DATA, @ MODEL

assembly language features

Identifiers:

- An identifier (or symbol) is a name that applies to an item in the program that expects to reference.
- Two types of identifiers are Name and Label.
 - Name refers to the address of a data item such as NUM1 DB 5, COUNT DB 0
 - Label refers to the address of an instruction.
- E. g: MAIN PROC FAR
- L1: ADD BL, 73

assembly language features

Statements:

- ALP consists of a set of statements with two types
- **Instructions**, e. g. MOV, ADD
- **Directives**, e. g. define a data item

	Identifiers	operation	operand	comment
Directive:	COUNT	DB	1	; initialize count
Instruction:	L30:	MOV	AX, 0	; assign AX with 0

assembly language features

Directives:

- The directives are the number of statements that enables us to control the way in which the source program assembles and lists.
- These statements called directives act only during the assembly of program and generate no machine-executable code. The different types of directives are:

assembly language features

1) The page and title listing directives:

- The page and title directives help to control the format of a listing of an assembled program.
- This is their only purpose and they have no effect on subsequent execution of the program.
- The page directive defines the maximum number of lines to list as a page and the maximum number of characters as a line.
- PAGE [Length] [Width]
- Default : Page [50][80]
- TITLE gives title and place the title on second line of each page of the program.
- TITLE text [comment]

assembly language features

2) SEGMENT directive

- It gives the start of a segment for stack, data and code.

Seg-name Segment [align] [combine][‘class’]

Seg-name ENDS

- Segment name must be present, must be unique and must follow assembly language naming conventions.
- An ENDS statement indicates the end of the segment.
- Align option indicates the boundary on which the segment is to begin; PARA is used to align the segment on paragraph boundary.
- Combine option indicates whether to combine the segment with other segments when they are linked after assembly. STACK, COMMON, PUBLIC, etc are combine types.
- Class option is used to group related segments when linking. The class code for code segment, stack for stack segment and data for data segment.

assembly language features

3) PROC Directives

- The code segment contains the executable code for a program, which consists of one or more procedures, defined initially with the PROC directives and ended with the ENDP directive.
- PROC - name PROC [FAR/NEAR]
-
- PROC – name ENDP
- FAR is used for the first executing procedure and rest procedures call will be NEAR.
- Procedure should be within segment.

assembly language features

4) END Directive

- An END directive ends the entire program and appears as the last statement.
- ENDS directive ends a segment and ENDP directive ends a procedure.
- END PROC-Name

5) ASSUME Directive

- An .EXE program uses the SS register to address the stack, DS to address the data segment and CS to address the code segment.
- Used in conventional full segment directives only.
- Assume directive is used to tell the assembler the purpose of each segment in the program.
- Assume SS: Stack name, DS: Data Segname CS: code segname

assembly language features

6) Processor directive

- Most assemblers assume that the source program is to run on a basic 8086 level computer.
- Processor directive is used to notify the assembler that the instructions or features introduced by the other processors are used in the program.
- E.g. .386 - program for 386 protected mode.

assembly language features

7) Dn Directive (Defining data types)

➤ Assembly language has directives to define data syntax: [name] Dn expression

➤ The Dn directive can be any one of the following:

➤ DB Define byte 1 byte

➤ DW Define word 2 bytes

➤ DD Define double 4 bytes

➤ DF defined farword 6 bytes

➤ DQ Define quadword 8 bytes

➤ DT Define 10 bytes 10 bytes

VAL1 DB 25

ARR DB 21, 23, 27, 53

MOV AL, ARR [2] or

MOV AL, ARR + 2 ; Moves 27 to AL register

assembly language features

8) The EQU directive

- It can be used to assign a name to constants.
- E.g. `FACTOR EQU 12`

9) DUP Directive

- It can be used to initialize several locations to zero. e. g. `SUM DW 4 DUP(0)`
- Reserves four words starting at the offset sum in DS and initializes them to Zero.
- Also used to reserve several locations that need not be initialized. In this case (?) is used with DUP directives. E. g. `PRICE DB 100 DUP(?)`
- Reserves 100 bytes of uninitialized data space to an offset PRICE.

assembly language features

10) DOSSEG

- There is a standard order for placing the stack, code and data segments One can place the segments in any order as well.
- This directives tells the assembler to place the segments in standard order

11) MODEL

- This directive determines the size of each segment .All of the program models except tiny result in the creation of exe program. The tiny model creates a com program.

assembly language features

Model	Description
TINY	Code and data together may not be greater than 64K
SMALL	Neither code nor data may be greater than 64K
MEDIUM	Only the code may be greater than 64K
COMPACT	Only the data may be greater than 64K
LARGE	Both code and data may be greater than 64K
HUGE	All available memory may be used for code and data

assembly language features

ORG:

- It changes the starting offset address of the data.

STARTUP

- It generates the instruction to initialize the segment register

EXIT

- It generates the INT 21H function, 4CH instruction for exiting program.

Global:

- The GLOBAL directive is used to make the symbol available to the other modules.

EXTERN:

- It is used to tell the assembler that the name or label following the directive are in some other assembly module

Microprocessor

1

In 8085 microprocessor, how many interrupts are maskable.

- a. Two
- b. Three
- c. Four
- d. Five

2

In the instruction of the 8085 microprocessor, how many bytes are present?

- a. One or two
- b. One, two or three
- c. One only
- d. Two or three

3

Which one of the following register of 8085 microprocessor is not a part of the programming model?

- a. Instruction register
- b. Memory address register
- c. Status register
- d. Temporary data register

Microprocessor

4

Which stack is used in 8085 microprocessors?

- a. FIFO
- b. FILO
- c. LIFO
- d. LILO

5

Which one of the following addressing technique is not used in 8085 microprocessor?

- a. Register
- b. Immediate
- c. Register indirect
- d. Relative

6

The program counter in 8085 microprocessor is a 16-bit register, because

- a. It counts 16 bits at a time
- b. There are 16 address times
- c. It facilitates the users storing 16-bit data temporarily
- d. It has to fetch two 8-bit data at a time.

Microprocessor

7

A direct memory access (DMA) transfer replies

- a. Direct transfer of data between memory and accumulator
- b. Direct transfer of data between memory and I/O devices without the use of microprocessor
- c. Transfer of data exclusively within microprocessor registers
- d. A fast transfer of data between microprocessor and I/O devices

8

In a Microprocessor, the address of the new next instruction to be executed is stored in

- a. Stack pointer
- b. address latch
- c. Program counter
- d. General purpose register

9

The instruction JNC 16-bit refers to jump to 16-bit address if ?

- a. sign flag is set
- b. carry flag is reset
- c. zero flag is set
- d. parity flag is reset

Microprocessor

10

The microprocessor of a computer can operate on any information if it is present in _____ only.

- a) Program Counter
- b) Flag
- c) Main Memory
- d) Secondary Memory

11

Which one of the following statements is correct regarding the instruction CMP A ?

- a. compare accumulator with register A
- b. compare accumulator with memory
- c. compare accumulator with register H
- d. This instruction does not exist

12

XCHG instruction of 8085 exchanges the content of ?

- a. top of stack with contents of register pair
- b. BC and DE register pairs
- c. HL and DE register pairs
- d. None of the above

Microprocessor– SET A

13 Direction flag is used with

- a. string instructions
- b. stack instructions
- c. arithmetic instructions
- d. branch instructions

14 The register which holds the information about the nature of results of arithmetic or logic operations is called as

- a. Accumulator
- b. Condition code register
- c. Flag register
- d. Process status registers

15 Which of the following interrupt is non-vectored in 8085?

- ☐ RST 5.5
- ☐ TRAP
- ☒ INTR
- ☐ RST 7

Microprocessor– SET A

16 Following is a 16-bit register for 8085 microprocessor

- ☒ a. Stack pointer
- b. Accumulator
- c. Register B
- d. Register C

17 When referring to instruction words, a mnemonic is

- a. a short abbreviation for the operand address.
- ☒ b. a short abbreviation for the operation to be performed.
- c. a short abbreviation for the data word stored at the operand address.
- d. Shorthand for machine language.

18 . What kind of interrupts are RST0 to RST7 in the 8085 microprocessor?

- ☐ Logical interrupts
- ☐ Hardware interrupts
- ☐ Conditional interrupts
- ☒ Software interrupts

Microprocessor

19

Which of the following is true about microprocessors?

- a) It has an internal memory
- b) It has interfacing circuits
- c) It contains ALU, CU, and registers
- d) It uses Harvard architecture

20

The _____ directive instructs the assembler to begin memory allocation for a segment/block/code from the stated address.

- a) GROUP
- b) OFFSET
- c) ORG
- d) LABEL

21

Which of the following flag is used to mask INTR interrupt?

- a) zero flag
- b) auxiliary carry flag
- c) interrupt flag
- d) sign flag

Microprocessor

22

Which of the following is the correct sequence of operations in a microprocessor?

- a) Opcode fetch, memory read, memory write, I/O read, I/O write
- b) Opcode fetch, memory write, memory read, I/O read, I/O write
- c) I/O read, opcode fetch, memory read, memory write, I/O write
- d) I/O read, opcode fetch, memory write, memory read, I/O write

23

Which of the following is not a property of TRAP interrupt in microprocessor?

- a) It is a non-maskable interrupt
- b) It is of highest priority
- c) It uses edge-triggered signal
- d) It is a vectored interrupt

24

Which of the following circuit is used as a special signal to demultiplex the address bus and data bus?

- a) Priority Encoder
- b) Decoder
- c) Address Latch Enable
- d) Demultiplexer

Microprocessor

25

How many flip-flops are there in a flag register of 8085 microprocessor?

a) 4

b) 5

c) 7

d) 10

26

Whenever a non-maskable interrupt occurs in 8085 microprocessor, which of the following data line contains the data?

a) 2C H

b) 3C H

c) 36 H

d) 24 H

27

How many address lines are present in 8086 microprocessor?

a) 16

b) 20

c) 32

d) 40

Microprocessor

28 Which of the following flag condition is used for BCD arithmetic operations in microprocessor?

- a) Sign flag
- b) Auxiliary carry flag**
- c) Parity flag
- d) Zero flag

29 What does a microprocessor understand after decoding opcode?

- a) Perform ALU operation
- b) Go to memory
- c) Length of the instruction and number of operations**
- d) Go to the output device

30 Which of the following is not a status flag in microprocessor?

- a) Overflow flag
- b) Direction flag
- c) Interrupt flag
- d) Index flag**

Microprocessor

31

Which of the following is not a condition flag?

- a) Trap flag
- b) Auxiliary carry flag
- c) Parity flag
- d) Zero flag

32

A memory connected to a microprocessor has 20 address lines and 16 data lines. What will be the memory capacity?

- a) 8 KB
- b) 2 MB
- c) 16 MB
- d) 64 KB

33

Which of the following is not true about 8085 microprocessor?

- a) It is an 8-bit microprocessor
- b) It is a 40 pin DIP chip
- c) It is manufactured using PMOS technology
- d) It has 16 address lines

Microprocessor

34

Which of the following register is not used in opcode fetch operations?

- a) Program counter
- b) Memory address register
- c) Memory data register
- d) Flag register

35

What is the word length of the Pentium-II microprocessor?

- a) 8-bit
- b) 32-bit
- c) 64-bit
- d) 16-bit

36

Which of the following is true?

- a) Every instruction has two parts i.e. opcode and operands
- b) MOV B, C is a two-byte instruction
- c) MVI A, 90H is a three-byte instruction
- d) Maximum number of T-states possible for the execution of an instruction is 16

Microprocessor

37

If a 90 GB memory has to be connected to a microprocessor, minimum how many address lines are required?

- a) 36
- b) 39
- c) 32
- d) 37

38

Which of the following is true about stack pointer?

- a) Stack pointer contains the address of the top of the stack memory
- b) Stack pointer is an 8-bit register
- c) Stack pointer stores data permanently
- d) Stack pointer is initialized after stack operation

39

Which of the following is true about MOV A, B instruction?

- a) It means move the content of register A to register B
- b) It uses immediate addressing mode
- c) It doesn't affect the flag register
- d) It is a 2-byte instruction

Microprocessor

40

Which of the following is a software interrupt?

- a) TRAP
- b) INTR
- c) RST-6.5
- d) RST-5

41

How many address lines are required to connect a 4 KB RAM to a microprocessor?

- a) 10
- b) 16
- c) 12
- d) 20

42

Which of the following is false about LDA instruction?

- a) It is a 3-byte instruction
- b) It uses indirect addressing mode
- c) It has 13 T-states
- d) It doesn't affect any flags

Microprocessor

43

DAA instruction is used to perform which type of addition?

- a) BCD addition
- b) Excess-3 addition
- c) Binary addition
- d) Octal addition

44

_____ converts the programs written in assembly language into machine instructions.

- a) Machine compiler
- b) Interpreter
- c) Assembler
- d) Converter

45

The assembler directive EQU, when used in the instruction: Sum EQU 200 does _____

- a) Finds the first occurrence of Sum and assigns value 200 to it
- b) Replaces every occurrence of Sum with 200
- c) Re-assigns the address of Sum by adding 200 to its original address
- d) Assigns 200 bytes of memory starting the location of Sum

Microprocessor

46

Suppose registers 'A' and 'B' contain 50H and 40H respectively. After instruction MOV A, B, what will be the contents of registers A and B?

a) 40H, 40H

b) 50H, 40H

c) 50H, 50H

d) 60H, 40H

47

The instructions like MOV or ADD are called as _____

a) OP-Code

b) Operators

c) Commands

d) None of the mentioned

48

To overcome the problems of the assembler in dealing with branching code we use ____

a) Interpreter

b) Debugger

c) Op-Assembler

d) Two-pass assembler

Microprocessor

49

Which of the following will be done in an 8085 microprocessor when an instruction LXI H 2070H is executed ?

1. 70 H is loaded in H register and 20 H is loaded in L register.
2. Content of the memory 2070H is loaded in H register.
3. 20 H is loaded in H register and 70 H is loaded in L register.
4. 2070H is loaded in H register.

50

The main purpose of Accumulator register of 8085 is

1. temporary data storage
2. selection of peripheral
3. storing instructions
4. used as primary pointer

Microprocessor

51

The 8085 Microprocessor has

1. 8 - bit data bus 16 - bit address bus

2. 8 - bit data bus 8 - bit address bus

3. 16 - bit data bus 8 - bit address bus

4. 16 - bit data bus 16 - bit address bus

52

How many bit Program counter is available in 8085?

1. 8 bit

2. 16 bit

3. 32 bit

4. 4 bit

53

In a microprocessor, the address of the next instruction to next executed, is sorted in

1. Stack pointer

2. Address latch

3. Program counter

4. General purpose register

54

What is the maximum addressing capability of Intel 8085?

1. 64 KB

2. 1 MB

3. 4 KB

4. 32 KB

Microprocessor

55

The clock frequency of an 8085 microprocessor is 5 MHz. If the time required to execute an instruction is $1.4 \mu\text{s}$, then the number of T-states needed for executing the instruction is

- 1. 1
- 2. 6
- 3. 7
- 4. 4

56

What is the length of SP (stack pointer)?

- 1. 6 bits
- 2. 8 bits
- 3. 12 bits
- 4. 16 bits

57

In a microprocessor, the term 'pipelining' refers to

- 1. address decoding
- 2. memory writes
- 3. fetching next instruction while the current instruction is executed
- 4. virtual memory addressing

58

In 8085 microprocessors, which signal is used to insert wait?

- 1. READY
- 2. ALE
- 3. HOLD
- 4. INTR

Microprocessor

59

The work of EU is _____

- A. encoding
- B. decoding**
- C. processing
- D. calculations

61

The register AX is formed by grouping _____

- A. AH & AL**
- B. BH & BL
- C. CH & CL
- D. DH & DL

60

The 16 bit flag of 8086 microprocessor is responsible to indicate _____

- A. the condition of result of ALU operation**
- B. the condition of memory
- C. the result of addition
- D. the result of subtraction

62

The microprocessor determines whether the specified condition exists or not by testing the _____

- A. carry flag
- B. conditional flag**
- C. common flag
- D. sign flag

THANK YOU